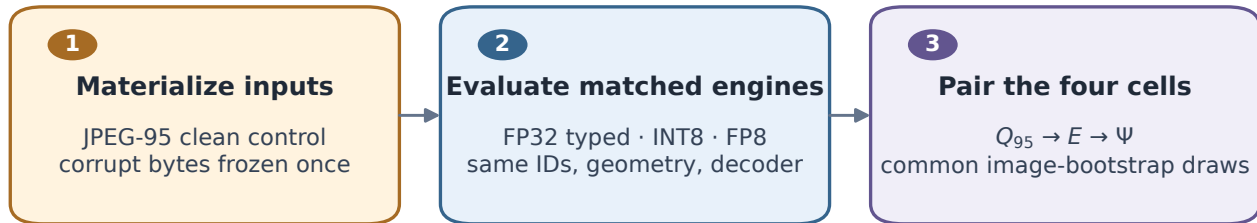


# Paired excess-gap evaluation framework



Direct format contrast:  $\Delta E = E_{\text{INT8}} - E_{\text{FP8}}$ ; FP32 cancels |  $\Delta \Psi = \Psi_{\text{INT8}} - \Psi_{\text{FP8}}$

Shared bytes and image resamples; SHA-256: checkpoint → engine → input → metric → paired-bootstrap artifact